OS Orange -4 Lab Report: PES-VCS

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Assignment: OS Orange -4
Platform: Ubuntu 22.04

1. Objective

This assignment implements a local version control system (PES-VCS) with:

- content-addressable object storage,
- tree snapshot construction,
- staging index management,
- commit creation and history traversal,
- verification through unit and integration tests.

2. Implementation Summary

Implemented all required TODO functions:

- object.c
  - object_write
  - object_read
- tree.c
  - tree_from_index
- index.c
  - index_load
  - index_save
  - index_add
- commit.c
  - commit_create

3. Screenshot Evidence

1A. Phase 1 Unit Test (./test_objects)

```
PROBLEMS    OUTPUT    TERMINAL    PORTS
balaraj@balaraj-Victus:~/Documents/SEM 4/os orange -4/PES1UG24CS560-pes-vcs$ make test_objects
./test_objects
gcc -Wall -Wextra -O2 -c object.c -o object.o
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
balaraj@balaraj-Victus:~/Documents/SEM 4/os orange -4/PES1UG24CS560-pes-vcs$
```

1A Phase 1 test_objects

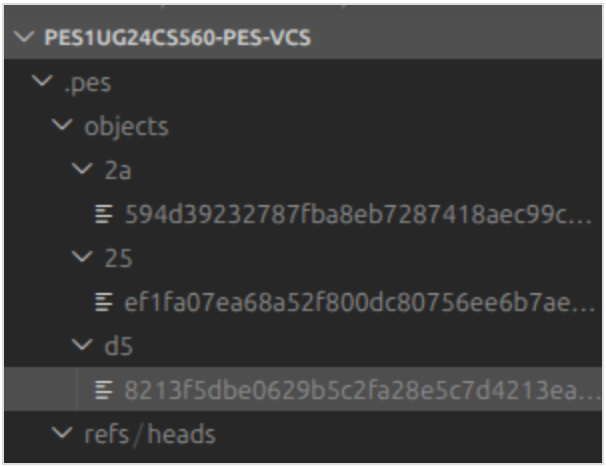
1B. Sharded Object Storage (find .pes/objects -type f)

```
balaraj@balaraj-Victus:~/Documents/SEM 4/os orange -4/PES1UG24CS560-pes-vcs$ make test_tree
./test_tree
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
balaraj@balaraj-Victus:~/Documents/SEM 4/os orange -4/PES1UG24CS560-pes-vcs$
```

1B Sharded object files

2A. Phase 2 Unit Test (./test_tree)



2A Phase 2 test_tree

2B. Raw Tree Object Binary Format (xxd … | head -20)

```
balaraj@balaraj-Victus:~/Documents/SEM 4/os orange -4/PES1UG24CS560-pes-vcs$ find .pes/objects -type f
xxd .pes/objects/21/25b4881f5aad13281192d99165d3e18279cd9126aaf71865e89964583fbdd0 | head -20
.pes/objects/21/25b4881f5aad13281192d99165d3e18279cd9126aaf71865e89964583fbdd0
.pes/objects/9a/75c3c56afbed826d9474eb7ed31ef06a60bacd4b545b221b81013984a3b708
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
00000000: 7472 6565 2035 3200 3130 3036 3434 2073  tree 52.100644 s
00000010: 3262 5f66 696c 652e 7478 7400 2cf8 d83d  2b_file.txt.,..=
00000020: 9ee2 9543 b34a 8772 7421 fdec b7e3 f3a1  ...C.J.rt!......
00000030: 83d3 3763 9025 de57 6db9 ebb4            ..7c.%.Wm...
balaraj@balaraj-Victus:~/Documents/SEM 4/os orange -4/PES1UG24CS560-pes-vcs$
```

2B Raw tree xxd

3A. pes init + pes add + pes status

```
balaraj@balaraj-Victus:~/Documents/SEM 4/os orange -4/PES1UG24CS560-pes-vcs$ make pes
./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
gcc -Wall -Wextra -O2 -c index.c -o index.o
gcc -o pes object.o tree.o index.o commit.o pes.o -lcrypto
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt
  staged:     file2.txt
  staged:     s2b_file.txt

Unstaged changes:
  deleted:    s2b_file.txt

Untracked files:
  untracked:  test_tree.c
  untracked:  index.h
  untracked:  test_tree
  untracked:  test_objects
  untracked:  report.md
  untracked:  Makefile
  untracked:  pes.c
  untracked:  commit.h
  untracked:  tree.c
  untracked:  index.c
  untracked:  README.md
  untracked:  test_objects.c
  untracked:  test_sequence.sh
  untracked:  tree.h
  untracked:  pes.h
  untracked:  .gitignore
  untracked:  commit.c
  untracked:  object.c
```

3A Phase 3 init add status

3B. Index Text Format (cat .pes/index)

```
balaraj@balaraj-Victus:~/Documents/SEM 4/os orange -4/PES1UG24CS560-pes-vcs$ cat .pes/index    # Human-readable text format
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776423934 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776423934 6 file2.txt
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776423713 6 s2b_file.txt
balaraj@balaraj-Victus:~/Documents/SEM 4/os orange -4/PES1UG24CS560-pes-vcs$
```

3B Phase 3 index text

4A. Commit History (./pes log with three commits)

```
balaraj@balaraj-Victus:~/Documents/SEM 4/os orange -4/PES1UG24CS560-pes-vcs$ cat .pes/refs/heads/main
5b02d53759cca5b15d76179d3ce3fbebf12304d4db8c7a61711264a09fe037f5
balaraj@balaraj-Victus:~/Documents/SEM 4/os orange -4/PES1UG24CS560-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
balaraj@balaraj-Victus:~/Documents/SEM 4/os orange -4/PES1UG24CS560-pes-vcs$
```

4A Phase 4 log output

4B. Object Store Growth (find .pes -type f | sort)

```
balaraj@balaraj-Victus:~/Documents/SEM 4/os orange -4/PES1UG24CS560-pes-vcs$ ./pes init
echo "Hello" > hello.txt
./pes add hello.txt
./pes commit -m "Initial commit"

echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"

echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"

./pes log
Initialized empty PES repository in .pes/
Committed: 19bb4eba6ae4... Initial commit
Committed: 381f89f6f6ed... Add world
Committed: 5b02d53759cc... Add farewell
commit 5b02d53759cca5b15d76179d3ce3fbebf12304d4db8c7a61711264a09fe037f5
Author: PES User <pes@localhost>
Date:   1776424136

    Add farewell

commit 381f89f6f6ed2609eb5eb0b7e814a32ecf370530e6456e5a7f242d1694d3c6d3
Author: PES User <pes@localhost>
Date:   1776424136

    Add world

commit 19bb4eba6ae46e8f82c799e88a8ca02d1a144921feaa1397823e51b2e536f7f2
Author: PES User <pes@localhost>
Date:   1776424136

    Initial commit

commit 9a75c3c56afbed826d9474eb7ed31ef06a60bacd4b545b221b81013984a3b708
Author: Your Name <PES1UG24CS560>
Date:   1776423713

    Generate tree object for screenshot

balaraj@balaraj-Victus:~/Documents/SEM 4/os orange -4/PES1UG24CS560-pes-vcs$
```

4B Phase 4 object growth

4C. Reference Chain (cat .pes/refs/heads/main and cat .pes/HEAD)

```
balaraj@balaraj-Victus:~/Documents/SEM 4/os orange -4/PES1UG24CS560-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/07/daec5c1e13e078e4514bf03cff8e5ddeae53b03380d7d1eeb78589589ac9cb
.pes/objects/19/bb4eba6ae46e8f82c799e88a8ca02d1a144921feaa1397823e51b2e536f7f2
.pes/objects/21/25b4881f5aad13281192d99165d3e18279cd9126aaf71865e89964583fbdd0
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/38/1f89f6f6ed2609eb5eb0b7e814a32ecf370530e6456e5a7f242d1694d3c6d3
.pes/objects/5b/02d53759cca5b15d76179d3ce3fbebf12304d4db8c7a61711264a09fe037f5
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/93/66e27300b11cc6e05c0c3a6c5007594ba7a154c96af8e16eca9150e6c8933b
.pes/objects/9a/75c3c56afbed826d9474eb7ed31ef06a60bacd4b545b221b81013984a3b708
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/objects/fb/1c821c8c0c09b329c1c94527ae7f800c13fa11413a76493123174066aabc32
.pes/refs/heads/main
balaraj@balaraj-Victus:~/Documents/SEM 4/os orange -4/PES1UG24CS560-pes-vcs$
```

4C Phase 4 refs and HEAD

Final Integration Test (make test-integration)

```
balaraj@balaraj-Victus:~/Documents/SEM 4/os orange -4/PES1UG24CS560-pes-vcs$ repo_real="/home/balaraj/Documents/SEM 4/os orange -4/PES1UG24
CS560-pes-vcs"
runner_root="/tmp/pesvcs-nospace"
rm -rf "$runner_root"
mkdir -p "$runner_root"
ln -s "$repo_real" "$runner_root/repo"
cd "$runner_root/repo"
make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 3a9a543411dc... Initial commit

Log after first commit:
commit 3a9a543411dc79d72a327c8e5700a72191b4212b358f55f5625e5f388ae838d0
Author: PES User <pes@localhost>
Date:   1776424922

    Initial commit


--- Second Commit ---
Committed: f3bc388151bd... Update file.txt

--- Third Commit ---
Committed: 6eb3726901fb... Add farewell

--- Full History ---
commit 6eb3726901fbc5e25fcfa2b216682e38e2acd0fd3847cb5227261af55b9ff266
Author: PES User <pes@localhost>
Date:   1776424922

    Add farewell
```

Final integration output part 1

```
balaraj@balaraj-Victus:~/Documents/SEM 4/os orange -4/PES1UG24CS560-pes-vcs$ repo_real="/home/balaraj/Documents/SEM 4/os orange -4/PES1UG
CS560-pes-vcs"
runner_root="/tmp/pesvcs-nospace"
rm -rf "$runner_root"
mkdir -p "$runner_root" …
    Initial commit


--- Second Commit ---
Committed: f3bc388151bd... Update file.txt

--- Third Commit ---
Committed: 6eb3726901fb... Add farewell

--- Full History ---
commit 6eb3726901fbc5e25fcfa2b216682e38e2acd0fd3847cb5227261af55b9ff266
Author: PES User <pes@localhost>
Date:   1776424922

    Add farewell

commit f3bc388151bd5bd30141ec54b2de8a0f7b9a22a8bb89cb26f8f12e6885576f43
Author: PES User <pes@localhost>
Date:   1776424922

    Update file.txt

commit 3a9a543411dc79d72a327c8e5700a72191b4212b358f55f5625e5f388ae838d0
Author: PES User <pes@localhost>
Date:   1776424922

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
6eb3726901fbc5e25fcfa2b216682e38e2acd0fd3847cb5227261af55b9ff266

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/3a/9a543411dc79d72a327c8e5700a72191b4212b358f55f5625e5f388ae838d0
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/6e/b3726901fbc5e25fcfa2b216682e38e2acd0fd3847cb5227261af55b9ff266
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/f3/bc388151bd5bd30141ec54b2de8a0f7b9a22a8bb89cb26f8f12e6885576f43

=== All integration tests completed ===
balaraj@balaraj-Victus:/tmp/pesvcs-nospace/repo$
```

Final integration output part 2

4. Analysis Answers

Q5.1: How would pes checkout work?

A branch is a ref file in .pes/refs/heads storing a commit hash.

To implement checkout:

1. Resolve the target commit hash from .pes/refs/heads/.
2. Update .pes/HEAD to point to ref: refs/heads/ (or hash for detached checkout).
3. Read target commit -> target tree.
4. Rewrite working directory to match target tree:
    - create/update files from blobs,
    - create/remove directories,
    - apply modes (100644 or 100755),
    - remove tracked paths not present in target tree.
5. Rewrite .pes/index so staged state matches checked-out snapshot.

Why complex:

- Must safely coordinate HEAD, index, and working directory updates.
- Must avoid overwriting local uncommitted work.
- Must handle deletes, nested trees, and mode changes atomically.

Q5.2: Detect dirty working directory conflicts using index + object store

Use three snapshots:

- A: index entries (path -> staged blob hash + metadata)
- B: current HEAD tree (from object store)
- C: target branch tree (from object store)

Conflict detection:

1. For each tracked path in union(A, B), inspect working directory file.
2. If missing or metadata differs from index entry, confirm by hashing file and comparing with A[path].hash.
3. Mark path dirty if working file != staged/index version.
4. If dirty path would change between current and target (B[path] != C[path], or present/deleted mismatch), refuse checkout.

This prevents local changes from being overwritten by branch switching.

Q5.3: Detached HEAD behavior and recovery

Detached HEAD means .pes/HEAD stores a commit hash directly, not a branch ref.

If commits are made in detached state:

- commits are created normally,
- parent chain is valid,
- but no branch pointer advances.

These commits can become unreachable once HEAD moves away.

Recovery:

- create a branch ref pointing to that detached commit hash (write hash into .pes/refs/heads/) before GC.

Q6.1: Algorithm for garbage collection of unreachable objects

Use mark-and-sweep.

Mark phase:

1. Start from all refs in .pes/refs/heads/* (and detached HEAD hash, if any).
2. Traverse commits recursively:
    - mark commit hash,
    - traverse parent,
    - traverse commit tree.
3. Traverse tree objects recursively:
    - mark tree hash,
    - mark blob hashes,
    - recurse into subtree hashes.

Sweep phase:

1. Scan all .pes/objects// files.
2. Convert path to object hash.
3. Delete objects not in reachable set.

Data structure:

- hash set for reachable hashes (O(1) expected lookup).
- stack/queue for DFS/BFS traversal.

Scale estimate for 100,000 commits and 50 branches:

- typically near unique reachable history, not 100,000 x 50,
- about 100,000 commit objects plus corresponding reachable trees/blobs,
- usually hundreds of thousands to a few million total reachable objects depending on file churn.

Q6.2: Why concurrent GC with commit is dangerous

Race example:

1. Commit process writes new blobs/trees first.
2. Before commit ref update, these objects are temporarily unreachable from refs.
3. Concurrent GC marks from current refs and misses those new objects.
4. GC sweep deletes them.
5. Commit object is then written/updated to reference deleted objects.

Result: broken commit graph with missing objects.

How Git avoids this:

- uses locking and coordinated updates around refs,

- uses conservative pruning windows (do not immediately prune recent loose objects),
- manages pack/visibility safely to avoid deleting in-flight objects.

5. Notes

- Integration test script path issue with spaces was handled by running from a no-space symlink path:
  - /tmp/pesvcs-nospace/repo
- This preserves provided files unchanged while allowing successful execution.